Dataset Inspection Report: Employee Attrition Analysis

Prepared by: Edwin Sunday Umana

Role: Data Science Intern, AnalystLab Africa

Date: 06 August 2026 (Week 1)

Batch: D

1. Dataset Overview

Source: IBM HR Analytics – Employee Attrition & Performance (Kaggle)

Total Rows: 1,470 employee records

Total Columns: 35 variables (features)

Target Variable: Attrition (Yes / No)

2. Data Structure and Types

The dataset consists of a healthy mix of numerical and categorical variables:

Numerical Variables: Age, Daily--Rate, Distance-From-Home, Education, Hourly-Rate, Job-Level, Monthly-Income, Num-Companies-Worked, Percent-Salary-Hike, Total-Working-Years, Years-At-Company, and other tenure/satisfaction metrics.

Categorical Variables: Attrition, Business Travel, Department, Education Field, Gender, Job Role, Marital Status, and Over Time.

Note: The Employee-Number column acts as a unique identifier for each employee.

3. Data Quality: Missing Values and Duplicates

Missing Values: The dataset is exceptionally clean. An inspection of all 35 columns reveals zero (0) missing values. No data imputation techniques are required.

Duplicates: A check for exact duplicate rows returned zero (0) duplicates. Every employee record in the dataset is unique.

4. Descriptive Statistics Summary

Age Distribution: Employee ages range from 18 to 60 years, with an average (mean) age of approximately 36.9 years.

Income Distribution: Monthly Income is right-skewed, ranging from $1,009 to $19,999. The mean income is $6,502, but the median is notably lower ($4,934), indicating that the majority of the workforce earns in the lower-to-middle income brackets.

Tenure: Years-At-Company ranges from 0 to 40 years, with an average tenure of 7 years.

Commute: Distance-From-Home ranges from 1 to 29 miles, with an average commute of 9.1 miles.

5. Initial Observations & Feature Redundancy

While the data is clean regarding nulls and duplicates, the inspection revealed a few constant/redundant columns that hold no predictive value:

Employee-Count (Every row is exactly 1)

Over18 (Every row is exactly 'Y')

Standard-Hours (Every row is exactly 80)

Conclusion

The dataset is of high quality and ready for Exploratory Data Analysis (EDA). However, the redundant columns mentioned above, along with the Employee-Number identifier, should be dropped before building any predictive machine learning models to prevent data leakage and reduce noise. Additionally, the target variable (Attrition) is imbalanced (approx. 16% Yes vs 84% No), which will need to be addressed during the future modelling phase.